Curve shortlist: 46 to 59-bit ECDLP targets and deep isogeny-volcano curves (2026-10-16)

Question and scope

The owner asked for two sets of curves:

1. One binary, one Koblitz and one prime-field curve whose prime subgroup is at least 46 bits, ideally in the 50s.
2. Curves with unusually deep isogeny-volcano structure: many levels, a large conductor gap between the crater (End = O_K) and Z[π], and several volcanoes at once.

This is a **curve search and exact construction**. It contains no ECDLP measurement, no speedup claim and no change to any scoreboard. Every number below is exact output of PARI/GP. Each statement is labelled **derived** (exact computation in `shortlist.gp`), **native cross-check** (agrees with the repository's Rust `curve_traits` engine) or **proposed** (a use the curves are offered for, not tested).

Method

- `shortlist.gp` (PARI/GP 2.15.4, `gp -q research/curve_shortlist_20261016/shortlist.gp`, about 20 s) builds every curve, writes `curves.json` (one record per curve, which the registry builder harvests) and `searches.jsonl` (all search rows, negatives included). Everything is deterministic. Seeds are SHA-256 strings hashed with the shell's `sha256sum`. Each generator is the cofactor times the point with the least x-coordinate (integer encoding), using the first ordinate PARI returns.
- The first pass ran the same PARI 2.17 library through Python bindings in a scratch directory. That pass is not committed, as the "no Python" rule requires (AGENTS.md). Running `shortlist.gp` under GP 2.15 reproduces the same curves, with the same coefficients for all six models that did not change field.
- **Volcano walks.** The script starts at the crater, a root of the Hilbert class polynomial `polclass(D_K)` or j = 0, and walks the modular polynomial `polmodular(ℓ)` without backtracking. A vertex's height above the floor is the length of the shortest non-backtracking walk to a degree-1 vertex. The descent takes one neighbour whose height is one less, then continues without backtracking.
- **Moduli** follow `docs/curves/ICV1.md`: the least sparse irreducible polynomial (`find_irreducible_sparse`) for GF(2^59), GF(2^61) and GF(2^73).
- **Registry.** All seven curves are registered (`docs/curves/registry.json`) with their EC1 representation (generator and subgroup). `traits.json`, `jacobian-transfers.json`, `covers.json` and the lab browser were regenerated from the new registry.

Part 1: three general targets (derived)

Curve	ICV1 slug	Field and model	#E	Prime subgroup n	log2 n
binary	`icv1-f2m59-tm247403969-f6f9b926`	GF(2^59), `x^59+x^7+x^4+x^2+1`; $y^2+xy = x^3+x^2+b$	2·n	288230376275413729	58.00
Koblitz (a = 0)	`icv1-f2m73-tm184271214331-9d25cc67`	GF(2^73), `x^73+x^4+x^3+x^2+1`; $y^2+xy = x^3+1$	4·27449·n	86020738150056119	56.26
prime	`icv1-fp56-tm252854635-0150ced1`	p = 2^56 − 5; $y^2 = x^3 - 3x + b$	n (prime order)	72057594290782567	56.00

- binary: b = `0x284697d4c7aba14` = SHA-256("binary59-15") mod 2^59. Seed index 15 is the first that gives #E = 2·prime.
- prime: b = 51345021292123855 = SHA-256("prime56-126") mod 2^56 mod p. Seed index 126 is the first that gives a prime #E. The twist order is 131 · 157 · 3503553935191.
- All three are ordinary. Every embedding degree is at least 2^49 (`embedding_degree` in `curves.json`), which rules out MOV/FR transfer. None is anomalous. The field degrees 59 and 73 are prime, so there is no subfield for Weil descent.
- **Koblitz negative search (derived, `searches.jsonl`):** for prime m in 46–70 the largest prime subgroup of any Koblitz curve is 48 bits (m = 61, a = 0). m = 73 is the first prime degree with a 50-to-59-bit prime factor. The Frobenius τ gives the usual √73 rho speedup on top of negation, so the Koblitz target is weaker than its 56 bits suggest.

Part 2: deep isogeny-volcano curves (derived)

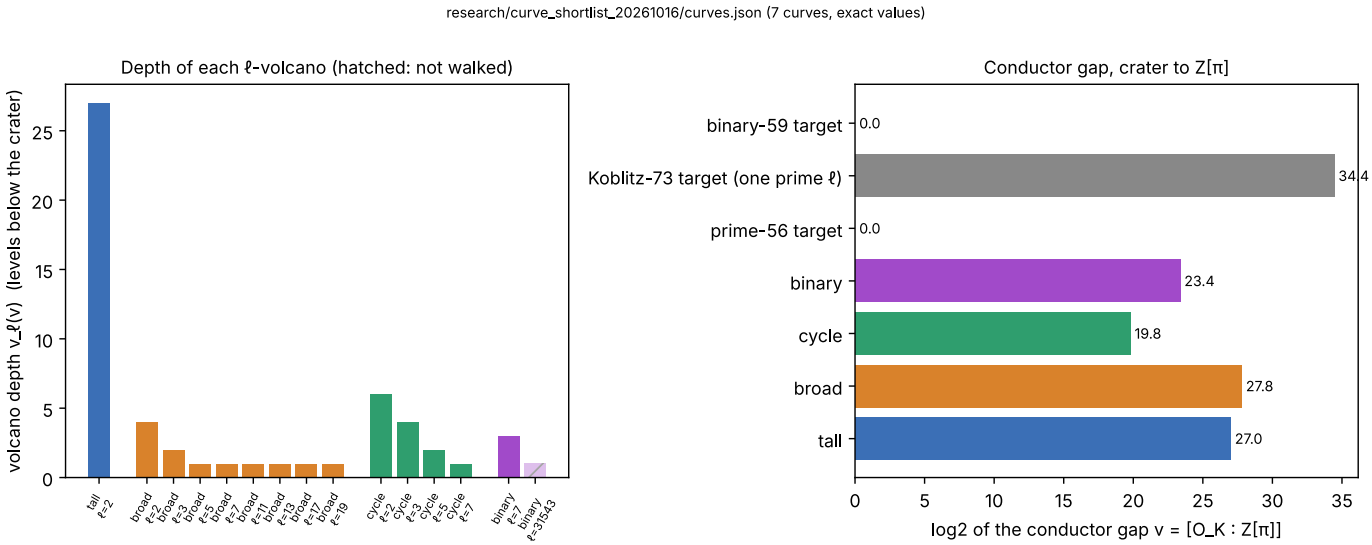
For an ordinary curve over F_q with trace t, $t^2 - 4q = v^2 \cdot D_K$. The ℓ-volcano has depth $v_\ell(v)$, and v is the conductor gap between O_K and Z[π]. The three targets above have v = 1, v = 1 and, for the Koblitz curve, the prime v = 23369856751: one huge single-level gap.

Curve	ICV1 slug	q	D_K (h_K)	v	Levels per ℓ (depth + 1)	#E	log2 n
tall	`icv1-fp56-tm300120038-865aea25`	p ≈ 2^55.0	−3 (1)	2^27	ℓ=2: 28	16·n	51.00
broad	`icv1-fp57-t354478198-afcea726`	p ≈ 2^56.0	−3 (1)	2^4·3^2·5·7·11·13·17·19	ℓ=2: 5, ℓ=3: 3, ℓ=5…19: 2 each	4·n	54.00
crater cycles	`icv1-fp56-t379549678-9040a9fd`	p ≈ 2^55.0	−71 (7)	2^6·3^4·5^2·7	ℓ=2: 7, ℓ=3: 5, ℓ=5: 3, ℓ=7: 2	4·n	53.00
binary	`icv1-f2m61-tm1864788689-c4a45971`	2^61, `x^61+x^5+x^2+x+1`	−49087 (122)	7^3·31543	ℓ=7: 4; ℓ=31543: 2 (not walked)	2·13·n	56.30

Where each curve sits, from the walk records in `curves.json`:

- **tall, broad, crater cycles: on the floor of every volcano.** The walk measured height 0 for every ℓ | v. So End(E) = Z[π].

  - tall: one 2-volcano with 28 levels. It has 2^27 vertices, and its floor level alone has 2^26 j-invariants. The crater is j = 0.
  - broad: eight volcanoes at once. $\ell = 3$ is ramified in $\mathbb{Q}(\sqrt{-3})$; $\ell = 7$, 13 and 19 split; $\ell = 2$, 5, 11 and 17 are inert. The walk took 12 descent steps.
  - crater cycles: $h(-71) = 7$. The craters for $\ell = 2$, 3 and 5 are 7-cycles: the script walked each one back to its start, with 2 crater neighbours at each vertex. $\ell = 7$ is inert. There are no extra automorphisms (units ±1). The crater is j = 21548212274226116.
- **binary: on the $\ell = 7$ floor but at the $\ell = 31543$ crater.** Φ_{31543} is out of reach, so the 31543-volcano was not walked. Hence $\mathrm{End}(E) = \mathbb{Z} + 7^3 \cdot O_K$, which strictly contains $\mathbb{Z}[\pi]$. A binary CM curve over a prime-degree field is rigid: m must divide $h(D_K)$, and v is then fixed by D_K. This curve has the largest small-ℓ depth among the 96 entries with $n \geq 50$ bits. Those come from a scan of 584 distinct (m, D_K, t) entries: 292 (m, D_K) pairs with $m \in \{53, 59, 61\}$, $|D_K| < 400000$ and $m \mid h(D_K)$.
- **Native cross-check.** `curve_traits show <slug>` independently derives, for all four curves, the same D_K, the same v (status `proved`), the same split/inert/ramified pattern and the same ℓ-volcano depths. That tool bounds $\mathrm{End}(E)$ between $\mathbb{Z}[\pi]$ and O_K. The exact level placement above comes from the GP walks.
- **Why the orders are even.** Depth at $\ell = 2$ forces t to be even, and with q odd that makes #E even. This is why tall, broad and crater cycles have 2-power cofactors. For odd $\ell \mid v$, $\ell \mid \#E$ only when $t \equiv 2 \pmod{\ell}$, and none of the chosen orders hits that.



Left: depth of each ℓ-volcano for the four deep curves (hatched = not walked). Right: log2 of v for all seven curves. Source: `curves.json`; the population is these seven curves, and every value is exact (no sampling uncertainty).

Generators, coefficients, order factorizations, crater j-invariants, per-ℓ heights at the crater and the floor, and descent step counts are all in `curves.json`.

Caveats and open checks

- **Hardness: proposed, not measured.** Each deep-volcano curve is joined to its crater by an explicit chain of small-prime-degree isogenies, so its ECDLP is no harder than the crater curve's. The crater of tall and broad is j = 0. Its order-6 automorphisms give rho an extra $\sqrt{3}$ on top of negation, so any cross-method comparison on these two must price rho with that speedup (AGENTS.md §5a). crater cycles and binary have only ±1 as automorphisms.
- **Height probe at j = 0.** At j = 0 the probe for $\ell = 3$ on broad reads 1 where the depth is 2 (`height_at_crater`). Multiple edges at j = 0 cause this. The $\ell = 3$ descent started from a non-special vertex, took 2 steps and ended at height 0, so the floor placement stands.
- **Not done:**
  - no Lean formalization: these are exact computations, not theorems;
  - no point-count certificate beyond PARI's `ellcard`, cross-checked by the native traits engine's order and trace;
  - the $\ell = 31543$ placement of the binary curve;
  - no ecbench or IC measurement on any of these curves.

Files

file	what
`shortlist.gp`	the whole search and construction; regenerates `curves.json` and `searches.jsonl`
`curves.json`	the seven curve records (registry input)
`searches.jsonl`	the Koblitz order table (m = 44–89) and every binary CM scan row
`plot.py`	lays out `curves.json` as `volcano_depths.svg` (plotting only)
`REPORT.typ`, `REPORT.pdf`	this report with the graph, as a PDF